

Assessment No. 02

Total Marks: 20

Submission Deadline

03-08-2026

Submission Instructions

- Submit **printed copy only**.
 - Include theory answers before code.
 - Paste complete Java source code.
 - Attach screenshots of output.
 - Clearly explain program logic using comments.
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Part A – Theory (10 Marks)

Q1. (2 Marks)

Explain Java Swing and Event-Driven Programming.

Q2. (2 Marks)

What is JDBC?

Explain the complete JDBC Architecture with a labeled diagram.

Q3. (3 Marks)

Discuss Multithreading in Java.

Explain

- Thread Lifecycle
- Thread Creation
- Synchronization

Q4. (3 Marks)

Explain Lambda Expressions.

Write advantages and practical applications.

Part B – Programming (10 Marks)

Question 1 (5 Marks)

Develop a **Student Registration System** using Java Swing.

The GUI should contain

- Student ID
- Name
- Department
- Semester

Buttons

- Save
- Clear
- Exit

Requirements

- Use JFrame
 - JPanel
 - JTextField
 - JLabel
 - JButton
 - ActionListener
 - Layout Manager
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Question 2 (5 Marks)

Develop a Java JDBC application connected with MySQL.

Create a table named **Students**

Implement

- Insert Record
- Update Record
- Delete Record
- Search Record

Use

- PreparedStatement
- Exception Handling
- Proper validation

Assessment Learning Outcomes

Students will demonstrate

- GUI Development
 - Event Handling
 - JDBC Connectivity
 - Database Operations
 - Java API Usage
 - Software Documentation
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